

Aziz Enam

Robotics & Machine Learning Engineer · New York, NY · azizenam@gmail.com · azizenam.github.io · @sazizenam

EXPERIENCE

Founding Member of Technical Staff, Alquist

May 2025 – Present · New York, NY

Architect, technical owner, and solo developer of an end-to-end onboard autonomous navigation stack for next-generation robotic platforms.

- Architected and deployed a vision-only autonomy pipeline spanning perception, localization, planning, and control.
- Deployed the onboard navigation stack on the NVIDIA Jetson Orin embedded platform, achieving sub-100 ms end-to-end latency from camera exposure to commanded velocity. The real-time pipeline integrated visual-inertial odometry (VIO), learned depth inference, truncated and Euclidean signed-distance field (TSDF/ESDF) mapping, wavefront cost maps, and a sampling-based model predictive control (MPC) policy.
- Developed a scalable automatic end-to-end metric 3D reconstruction pipeline for sensor calibration and generation of onboard localization maps. Engineered the system to operate robustly in sparse-feature environments, including warehouses and retail stores, enabling reconstruction of 50,000+ sq ft facilities and delivery of localization-ready artifacts within hours.

Senior Applied Scientist — Localization, Calibration & Mapping, Cruise

Jan 2020 – Nov 2024 · San Francisco, CA

Core engineer of the Localization, Calibration and Mapping team, responsible for all things related to sensor intrinsic and extrinsic calibrations. Promoted from ML/Robotics Software Engineer to Senior Applied Scientist in July 2022.

- Solo and joint inventor on multiple patents related to sensor calibration (see below).
- Co-developed and researched a fully automated calibration system for diverse sensors, including cameras, lidar, radar, and IMUs, in unstructured environments. This system achieves complete calibration of new vehicles within 1 mile of driving in parking lots or garages.
- Authored and maintained online calibration monitors designed to track the gradual drift of extrinsic calibrations in vehicles and perform updates when deemed safe.

Software Engineer — Computer Vision, Postmates X (Serve Robotics)

Jan 2018 – Dec 2020 · San Francisco, CA

Core engineer of an early-stage team for Postmates' autonomous rover project, Serve.

- Designed processes to calibrate cameras with lidar for sensor fusion, along with visualization tools and quality metrics.
- Developed and deployed deep learning systems for object detection and scene segmentation.

Software Engineer — Machine Learning, FirstRain

Aug 2016 – Jul 2017 · San Mateo, CA

- Worked on algorithms to remove and filter content from FirstRain's corpus that was not considered business relevant.

PATENTS

Map-assisted target detection for sensor calibration US 12,522,223 B2 (2026), GM Cruise Holdings

Extrinsically calibrates and checks autonomous-vehicle sensors on-road by matching live sensor returns against static objects identified from a semantic map layer.

Continuous radar calibration check US 12,386,057 B2 (2025), GM Cruise Holdings

Continuously monitors radar-to-lidar alignment by tracking relative object positions over time, triggering corrective action when drift exceeds a threshold.

Drive-through calibration process US 12,221,119 B2 (2025), GM Cruise Holdings

Validates a vehicle's sensor calibration by driving through an environment mapped with known targets, confirming extrinsic parameters stay within bounds before autonomous operation.

SKILLS

Scientific Computing: C++17, Python, PyTorch / TensorFlow / TensorRT, OpenCV, PCL, Eigen, Ceres, scikit-learn, NumPy / SciPy, CUDA

Robotics: ROS, Linear Algebra, Convex Optimization, State Estimation, Deep Learning, Optical Flow, Sensor Calibration, SLAM, Kalman Filters, Model Predictive Control, 3DGS

EDUCATION

University of California, Irvine — *M.S. in Applied Mathematics*

2014 – 2016

University of California, San Diego — *B.S. in Electrical Engineering and Mathematics*

2010 – 2014